

Bangladesh Travel Guide: An Android Mobile Application to Travel in a better way.

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A Project Report Presented in Partial Fulfillment of the Requirements for the Degree of Bachelor
of Science in Information and Communication Engineering

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DECLARATION

This project report is submitted to the Department of Information and Communication Engineering, Noakhali Science and Technology University, Sonapur, Noakhali in partial fulfillment of the requirements for having the 4th Year 2nd Semester in ICE. This is also needed to certify that the project work is under the B.Sc. (Engineering.) course of the Information and Communication Engineering. So, I, here by, declare that this project report has not been submitted elsewhere for the requirement of any kind of degree, diploma or publication.

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ACCEPTANCE

This project report is submitted to the Department of Information and Communication Engineering, Noakhali Science and Technology University, Sonapur, Noakhali in partial fulfillment of the requirements for having the 4th Year 2nd Semester B.Sc. (Engineering) in ICE.

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ABSTRACT

In this project an android app for tourist guide developed. This android app provides guide people who have wish to visit Bangladesh. This application helps the tourists to find Bangladesh most of all attractive places where people wish to visit. The application users find the services category of different places name as Historical, Parks, Nature, Amusement museums under any division. User get services from server in real time system. In the server site user find the visiting place description, like how to go or where the place situated, contact information, each place visiting timing, entry ticket information etc. This project contains list of hotels with details information like contact information, hotel services etc. For this site, need to collect data and store them on server both touring places and hotels. This application show these data by Network API call. Besides some place service details does not available, at present, so these information will include in future when these information will be available by their authority. For the dynamic site user need their android device GPS service on .In the user site this android app provides the tourist with city map depending on its current location which is enabled by the android phone user. This site consists of emergency amenities like nearby hospitals, police stations, banks and restaurants etc.

For the tourist security the application contains emergency contact service which can be available either user use the application offline. Apart these feature the application provided another feature of weather forecast for five days. The app services are completely online based. The free google open weather API's are limited in use and get refreshed by itself. Bangladesh is prone to natural calamities, people who wish to visit Bangladesh can get weather forecast information for major cities from the application. The service is provided by the open weather map API. This android application is beneficial for the tourist's having no idea about the places they want to visit. The tourists and people shifting to new cities can get a better guidance of the places they want to visit.

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LIST OF ABBREVIATIONS

Serial Number	Short Form	ABBREVIATION
1	REST	Representational state transfer
2	API	Application programming Interface
3	JSON	JavaScript Object Notation
4	URL	Uniform Resource Locator
5	UI	User Interface
6	XML	Extensible Markup Language
7	SQL	Structured Query Language

CHAPTER 1

INTRODUCTION

1.1 Overview

Bangladesh is blessed with natural beauty which attracts the tourists. She has beautiful beaches as well as the longest beach in the world. It is proud of her beautiful heritage and historical relics. People get refreshment and enjoyment from these beauties. When people get free time they want to visit different places. Some people are wants to visit unknown places. Then they need to know how to go to the destination places, need to know the location directions. Sometimes tourists get sick in the visited places and need to take emergencies. So this application will help tourist to visit places in safe and provide emergency services. Location based Services offer many advantages. Tourist get nearby emergency services like police stations, nearby hospitals. With the help of GPS in phones we get these services. This app mainly focuses on help to those people who are normally thinking about spend their vacation and do not know which is good and how to travel these tourist places. For a tourist the app work as guide. So here available features of Google API will be provided with its functionalities. This system can be used to view the location in the map that the user wishes to reach. The usage of this application greatly reduces the time required to search for a place. The application also leads to quicker decision making with respect to places to visit.

1.2 Travel in Bangladesh: an example

Bangladesh is a land of scenic beauty. To make best use of the natural gift, the country has developed facilities of modern hotels, motels, rest houses, youth inns and restaurants, as well as modern modes of communication, are available at all tourist attractions. Some of the better known

spots are: Dhaka – The City of Mosques; Chittagong – Gateway to the Bay of Bengal; Cox’s Bazar – The Tourist Capital; Rangamati – The Heart of the Lake District; Kaptai – The Lake Town, Sylhet – Land of Two Leaves and a Bud; Sundarban– Home of the Royal Bengal Tiger; Bandarban – The Hilly Resort; Kuakata – Where the Sun Rises and Sets in the Sea; Mainamati – Seat of Lost Dynasties; Mahasthangarh – The Ancient Seat of Buddhist Rule; Rajshahi – The Silk Producing Centre; Natore – Place of Dighapatiay Palace; Paharpur – The Seat of Buddhist Learning; Dinajpur – Old Ornamental Temple Town; and Mymensing - The Heart of Bengal’s Folklore. Bangladesh Parjatan Corporation (BPC) is an autonomous government organization that caters to the needs of local and foreign tourists. BPC has tourist facilities at Dhaka, Cox’s Bazar, Chittagong, Rangpur, Rangamati, Rashahi, Bogra, Finajpur, Kuakata, Sylhet and Sundarban.

It organizes package tours, sightseeing trips and excursions. A number of private sector tour operators are also active in the country.

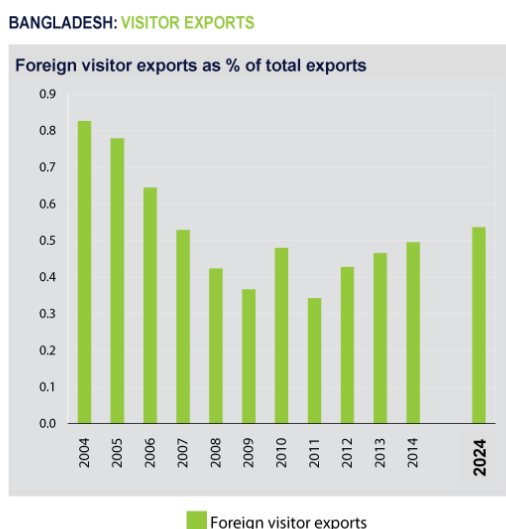


Figure 1.1: Visitor Exports
Tourism

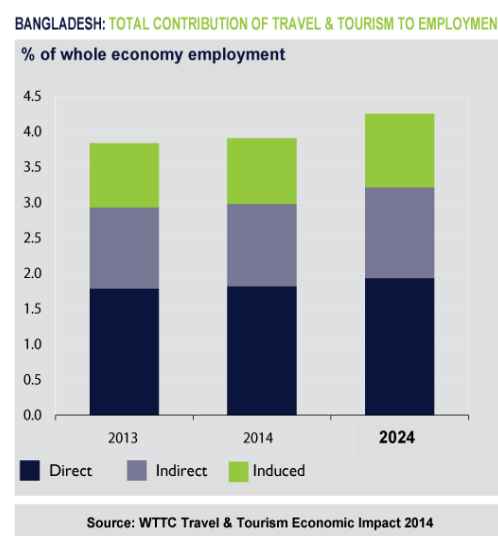


Figure 1.2: Total contribution of Travel &

1.3 Motivation

The proposed android app for travelling, hotel, nearby emergencies and weather forecast for some special reasons. Many travel agency in our country, they collect too much money from people because people do not know about these tourist places. But this project will be making sure the clear concept about these information. Anyone can see about all information about travelling, nearby emergencies and weather forecast. For that information they do not need to ask for any information. The app is developed so that no one get harassed by anyone when collection information and people also can know the history about tourist places accurately.

1.4 Objectives

Now we live in the era of Internet where internet make our life and fashion more easily. Information and communication technologies can improve access to data and quality of services while reducing the work and time. Proposed android system can be more manageable and safe for tourist who are feel comfort by using this application services and information.

The main objectives of the application are listed below:

- Provide reliable information.
- Tourist find tour places, all available information's in the application.
- Provide emergency information like nearby hospitals, police stations.
- People new in a place have find it efficient.
- This helps to give real time information speedy on the basis of user's current location, net speed.
- To develop our tourism sector.
- To utilize our free time in a meaningful way.
- Inform people about what amazing things happening around them.

1.5 Expected Outcome

This app is about travelling information, hotel information, nearby emergency services (hospitals, police stations, banks, and restaurant) and weather forecast. The people who want to spend their vacation or leisure time or want to travel, they can use our app for their all kinds of information about their tour. According to the demand of the user in future make this app more improvement.

1.6 Report Layout

The report is divided into six chapter where first chapter is combined with introduction, motivation, objectives and expected outcomes. The second chapter is included background of project, related works, and comparative studies, scope of the problem and challenges of the project. Third chapter is assembled with requirement specification which is includes business process modeling, requirement collection and analysis, use case modeling and description, logical data model and design requirements. In the chapter four discussed about design of the app of travelling, hotel, nearby services and weather forecast. In chapter 5, implement all and test all. In chapter six make a conclusion of all work and future scope which will do next.

CHAPTER 2

Background Study

2.1 Introduction

Travelling is the best way of refreshment. When people are busy with their work they need some refreshment. That's why they need to travel to some beautiful tourist places for their refreshment. They also want some kind of events for entertainment. They also need some rest in some special hotel in an unknown place. People need to collect information about these in some travel agencies. People need to weather forecast and nearby emergency humanities information. But for this work they have to cost time.

2.2 Related Works

In the recent work, I have found that some of similarities on these applications.

2.2.1 Travel Guide Bangladesh ^[1]

The application provides only Bangladesh tour place information.

2.2.2 Bangladesh Weather App ^[2]

The application ten days weather forecast for Bangladeshi major cities.

2.2.3 Travel Bangladesh ^[3]

This apps will provide description of the major tourist spot of Bangladesh. This website has some travel places.

2.2.4 Hotels of Bangladesh ^[4]

Find the correct information about popular and top ranked Hotels located in various tourist spots of Bangladesh. You will get phone numbers and hotel address at a glance. The website has some features such as hotel booking, hotel contacts.

2.3 Comparative Studies

In all these apps they have represented our country beautifully and our tourist places excellently. They have designed their app to attract the people around the country. Travel guide in Bangladesh application provide tour place information. Bangladesh weather application ten days weather forecast in major cities. Travel Bangladesh application provides tour places information only. These application provides different feature. But in the proposed app I gathered five features so that the user do not use four different app for travel, hotel information, weather forecast, apart these feature there added map feature to get nearby services like nearby restaurant, police stations, banks, hospitals and emergency call service system. They can get all information briefly by using the app.

2.4 Limitation of the Work

This app is simple including five features tour place information, hotel information, map services, weather forecast and emergency call service so that users can save their time by not looking for information in some travel agencies and have a safe tour. It has been difficult to collect some information about four features. User can also know the history of some tourist places using the developed app. But there is some limitations mainly the app impossible to use in offline. In future will developed to give service in offline to if map and weather forecast work offline.

2.5 Challenges

Important and advance feature of my project that was challenges to me such as:

1. Protection.
2. Security.
3. Real-time system.
4. Rest API
5. Live server data loading

CHAPTER 3

Methodology

3.1 Overview

Methodology is the systematic, theoretical analysis of the methods applied to a field of study. It comprises the theoretical analysis of the body of methods and principles associated with a branch of knowledge. Typically, it encompasses concepts such as paradigm, theoretical model, phases and quantitative or qualitative techniques.

3.2 Waterfall Model

For this application development waterfall model is followed as it is best suited to describe this project work step by step. The waterfall model was the first process model. It is also referred to as linear-sequential life cycle model which is shown in Figure 3.1. It is a sequential software development process, in which progress is seen as flowing steadily downwards through the phases of Conception, Initiation, Analysis, Design, Construction, Testing and Maintenance.

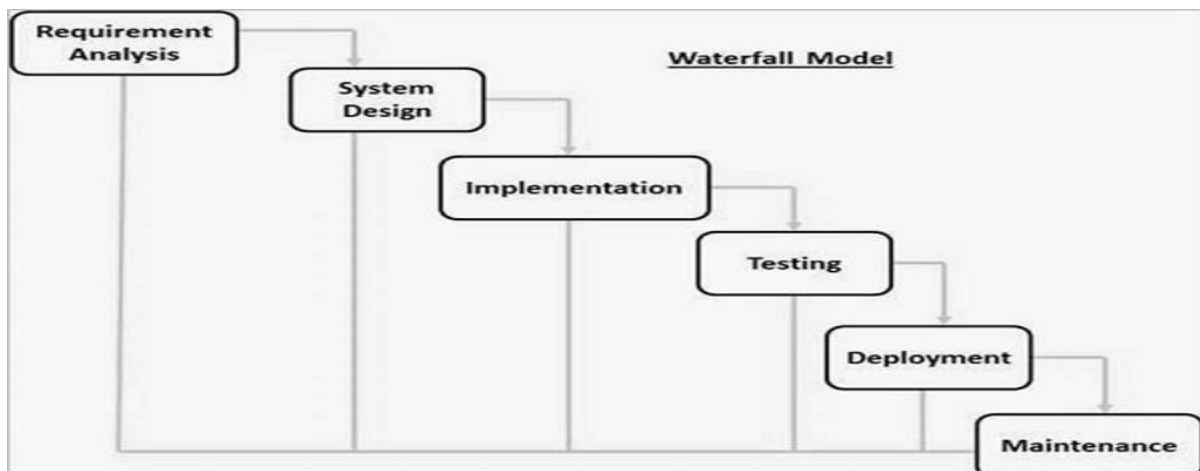


Figure 3.1: Waterfall Model.

3.3 Requirements specification

A software requirements specification (SRS) is a description of a software system to be developed. The software requirements specification lays out functional and non-functional requirements, and it may include a set of use cases that describe user interactions that the software must provide.

3.4 Systems design

At this step desired features and operations are described in detail, including screen layouts, business rules, process diagrams, pseudocode, and other documentation. The design phase involves the actual creation and design of a system.

3.5 Implementation

The objective of the Implementation Phase is: first to install the system in the production environment and to bring it into operation; and second, to ensure that the system, as developed to Solution Delivery Team.

3.6 Testing

After the code is developed it is tested against the requirements to make sure that the product is actually solving the needs addressed and gathered during the requirements phase. During this phase all types of functional testing like unit testing, integration testing, system testing, acceptance testing are done as well as non-functional testing are also done.

3.7 Deployment

After successful testing the product is delivered / deployed to the customer for their use. As soon as the product is given to the customers they will first do the beta testing. If any changes are required or if any bugs are caught, then they will report it to the engineering team. Once those changes are made or the bugs are fixed then the final deployment will happen.

3.8 Maintenance

Once when the customers starts using the developed system then the actual problems comes up and needs to be solved from time to time. This process where the care is taken for the developed product is known as maintenance.

3.9 Requirement Collection and Analysis

Developed this app, need to gathered many kinds of touring place list ^[5], place details ^[6], other information ^[7] and hotels information ^[8] from many different places and websites and analysis them and also saw some documents. Some information did not add because after analysis these are not validated. Many website does not provide their information for their business. Some of information are not available in google because authority does not want to provide.

CHAPTER 4

Design Specification

4.1 Front-end Design

Front-end design involves creating the XML code that makes up a user interface.

4.1.1 XML ^[9]

- XML stands for extensible Markup Language.
- XML was designed to store and transport data.
- XML was designed to be both human and machine-readable.

4.2 Back-end Design

4.2.1 JAVA ^[10]

Java allows you to play online games, chat with people around the world, calculate your mortgage interest, and view images in 3D, just to name a few. It's also integral to the intranet applications and other e-business solutions that are the foundation of corporate computing.

4.2.2 PHP ^[11]

PHP is a server scripting language, and a powerful tool for making dynamic and interactive Web pages. PHP is a widely-used, free, and efficient alternative to competitors such as Microsoft's ASP.

4.2.3 MySQL ^[12]

MySQL is a standard language for storing, manipulating and retrieving data in databases.

--SQL can retrieve data from a database

--SQL can insert, update, and delete records in a database

--SQL can create new databases

--SQL can create views in a database.

4.3 Implementation Requirements

Hardware Requirements:

- CPU 1.2 GHz Dual Core
- 512MB of RAM or higher.
- 19.46 MB of Storage space

Software Requirements

- Ice cream Sandwich (4.0.2) or higher.

4.4 Use case Modeling and Description

Figure 3.2 shows Use Case Modeling.

A use case (or set of use cases) has these characteristics:

- > Organizes functional requirements

- > Models the goals of system/actor (user) interactions.
- > Searches paths (called scenarios) from place title, place search, to goals.

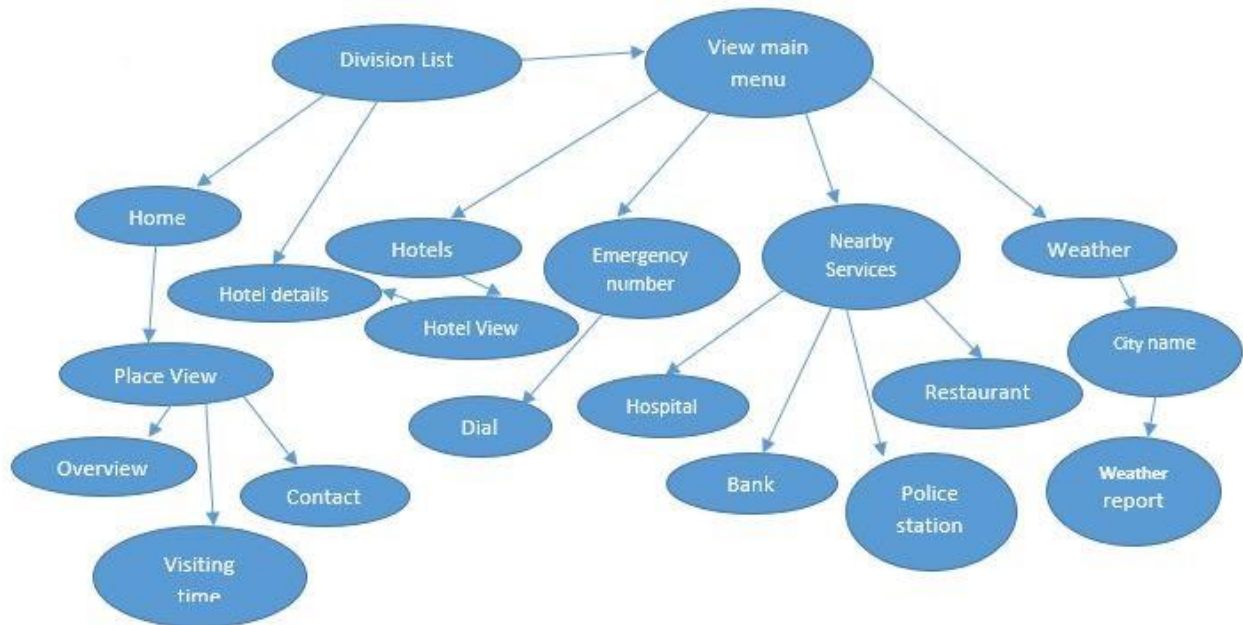


Figure 4.1: Use Case Modeling.

4.5 Logical Data Model

A logical data model describes how the system implemented.

4.5.1 Logical data model Schema

In here I am providing a schema which is shown in Figure 3.3, it is the structure for a database.

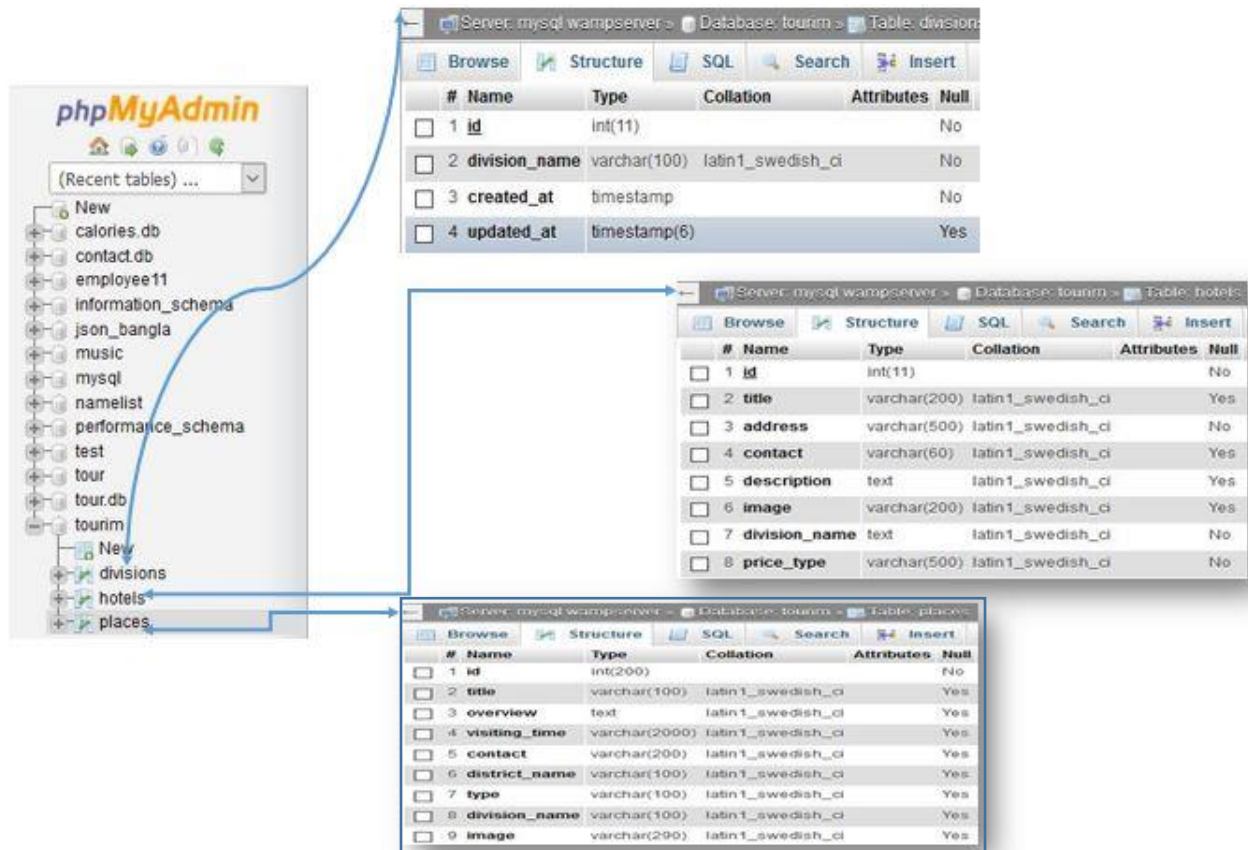


Figure 4.2: Logical Data Model (Schema).

4.6 Design Requirements

Design Requirements for admin:

- Take actions (edit/delete/view)
- Add features

Design Requirements for user:

- View
- Search
- GPS
- Internet

CHAPTER 5

Implementation

5.1 Implementation of Database

The implementation phase is where install the DBMS on the required hardware, optimize the database to run best on that hardware and software platform, and create the database and load the data. The initial data could be either new data captured directly or existing data imported from a MariaDB database or another DBMS. This establish database security in this phase and give the various users that identified access applicable to user requirements.

5.1.1 Main Table

Main Table contains places and place details information, hotel information which is shown in Figure 5.1. From here admin can edit these information. This table is main source of all data that used in this app.

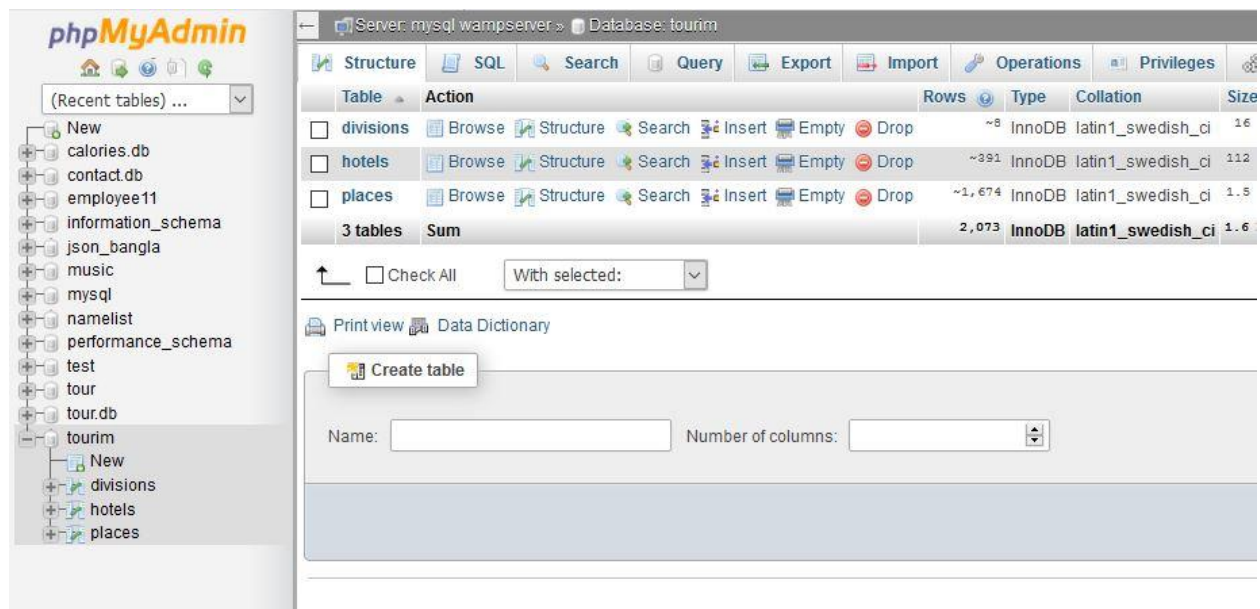


Figure 5.1: Main Table.

5.1.2 Divisions Table

This part is about divisions list which is shown in Figure 5.2. When a user want to select division wise tour place corresponding divisions are loaded from here and shown in navigation drawer. Divisions work as options to load all the tour places as division wise what is user requirement. So here this table is work for as an options to the corresponding application user.

					id	division_name	created_at	updated_at
☐					1	DHAKA	2018-01-22 02:18:20	NULL
☐					2	CHITTAGONG	2018-01-22 02:18:20	NULL
☐					3	BARISHAL	2018-01-22 02:18:43	NULL
☐					4	KHULNA	2018-01-22 02:18:43	NULL
☐					5	RAJSHAHI	2018-01-22 02:19:18	NULL
☐					6	RANGPUR	2018-01-22 02:19:18	NULL
☐					7	SYLHET	2018-01-22 02:19:34	NULL
☐					8	MYMENSINGH	2018-01-22 02:19:34	NULL

Figure 5.2: Divisions Table.

5.1.3 Places information Table

These tables are all about travelling which are shown in Figure 5.3 & 5.4. Users use these for getting information about places and hotels where they wants to stay. The application user get services each place details description, visiting time, contact, corresponding images and how to go where to stay. There are some lacking of information for few places because place authority does not provided the information. When get that information available these will be contained this table.



+ Options								
id	title	overview	visiting_time	contact	district_name	type	division_name	image
1	Ahsan Manzil	Islampur Road Sadarghat Dhaka - 3051 Ahsan Manz...	Gallery Timings: In Summer, you may visit the pala...		Dhaka	Landmarks	DHAKA	ahsanmanzil.jpg
2	Lalbagh Fort	Lalbagh Road Dhaka Lalbagh Fort is located at...	Opening Hours April to September (Summer) Tuesda...		Dhaka	Landmarks	DHAKA	lalbaghfort.jpg
3	Baitul Mukarram	Tophkhana Road Dhaka - 1000 Baitul Mukarram Mosq...			Dhaka	Landmarks	DHAKA	baitulmukarram.jpg
4	Hussaini Dalan	Bakshi Bazar Dhaka - 1211 Hussaini Dalan is lo...			Dhaka	Landmarks	DHAKA	hussainidalan.jpg

Figure 5.3: Places information Table.

5.1.4 Hotels Table

The table contains divisions category wise hotel choose options under in any divisions. User can select division wise also can be able to choose price wise like best values or lowest prices. These two selection option that is division wise or price wise are usage as search options. User find it more efficient cause the application contains all the information of all the hotels like address, contact, photo, services, food restaurant and price information.

tour.db

tourim

New

divisions

hotels

places

← →	▼	id	title	address	contact	description	image	division_name	price_type
Edit Copy Delete		1	Lakeshore Banani	Road 13A, House 81, Block D, Banani Dhaka, Dhaka...	01978-616164	Free Wifi Free Parking Breakfast included Air C...	lakeshorebanani.jpg	DHAKA	Best Value
Edit Copy Delete		2	The Westin Dhaka	Main Gulshan Avenue Plot-01, Road 45, Gulshan-2,...	02-9884800	Free Wifi Free Parking Breakfast Buffet Air Condit...	thewestindhaka.jpg	DHAKA	Best Value
Edit Copy Delete		3	Six Seasons Hotel	Main Gulshan Avenue Plot-01, Road 45, Gulshan-2,...		Free Wifi Free Parking Breakfast included Air C...	sixseasonshotel.jpg	DHAKA	Best Value
Edit Copy Delete		4	Le Meridien Dhaka	79/A Commercial Area Airport Road Nikunja 2, Khi...	02-8900089	Free Wifi Free Parking Breakfast included Air C...	lemeriendhaka.jpg	DHAKA	Best Value

Figure 5.4: Hotels Table.

5.2 Implementation of REST API

5.2.1 Implementation of weather forecast API ^[13]

In this section the API value of five days weather forecast that open weather provided that need to handle by java coding is mentioned. The data can be handled by two ways first one either JSON object or XML. The app used JSON object created model or POJO class, created java a class which return data as array list. Then requested to the API and properly handled to show all the data in the application user interface. Some of the functionality are not handled here in this project. Because this implementation needs more time and free open weather API does not provide all the services. A figure is given to see the weather API for five days that provided by google.

JSON	Raw Data	Headers
Save	Copy	Collapse All Expand All
cod:		"200"
message:		0
city:		
geoname_id:		524901
name:		"Moscow"
lat:		55.7522
lon:		37.6156
country:		"RU"
iso2:		"RU"
type:		"city"
population:		0
cnt:		7
list:		
0:		
dt:		1485766800
temp:		
day:		262.65
min:		261.41
max:		262.65
night:		261.41
eve:		262.65
morn:		262.65
pressure:		1024.53
humidity:		76
weather:		
0:		
id:		800
main:		"Clear"
description:		"sky is clear"
icon:		"01d"
speed:		4.57
deg:		225
clouds:		0

Figure 5.5: weather forecast API.

5.3 Implementation of API & Server Architecture

The architecture can be divided into 3 modules. These are:

- Getting location
- Database module
- Client Application

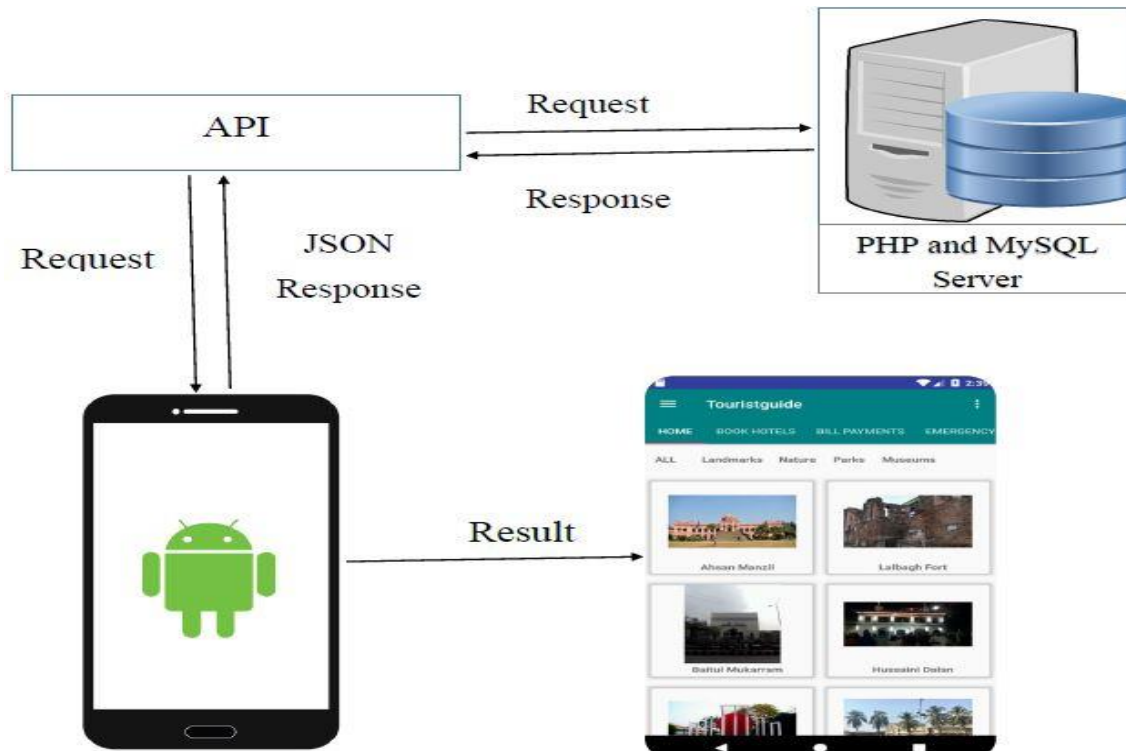


Figure 5.6: Procedure of loading data from database.

5.4 Implementation of front-end Design

This section contains the part that user shown which are known as user interfaces.

5.4.1 Main Features

When user use this App this interface will be shown to them which is shown in Figure 5.7. There are various option that they can choose. They can change division wise tour places also can filter each divisions tour places in historic, nature, parks, and museums category. User can select any

feature that they interested in. There is a search option that used for search by tour place name. This search option also work with search by hotels name.



Figure 5.7: Main Features.

5.4.2 Divisions List

When application initialized by user main UI comes with navigation drawer where contains all the divisions list.



Figure 5.8: Division list

5.4.3 Places Details

This is a figure of a tour place details which is shown in Figure 5.9. User see a tour place like this figure. The application when initialized by user they get tor places list. By clicking each list they get details of that tour place information like overview, visiting time and contact.



Figure 5.9: Places Details.

5.4.4 Hotel List

This is a figure of hotel list which is shown in Figure 5.10. Here is hotel search two options one is division wise another one is price category like Best value and lowest price. Apart these hotels can be search by division wise.



Figure 5.10: Hotel list.

5.4.5 Hotel Details

This is a figure of Hotel list details which is shown in Figure 5.11. Hotels will be shown here. Users can get all information about hotels especially whose are provided hotels contact information and other information available.



Figure 5.11: Hotel Details.

5.4.6 Emergency:

This section provide emergency call for sudden accident, harassment etc. for emergency help service.



Figure 5.12: Emergency services

5.4.7 Nearby emergency services using map.

This services need user their usage devices GPS on. When click each of services the services take their in google map and provided corresponding services.



Figure 5.13: Nearby Services

5.4.8 Navigation of nearby emergencies

This part shows the result of nearby emergencies humanities. Get this feature needs to user enter banks, hospitals, police stations and Restaurant like figure 5.14, 5.15, 5.16 and 5.17.

5.4.8.1. Nearby Banks

Navigation map view for nearby banks.

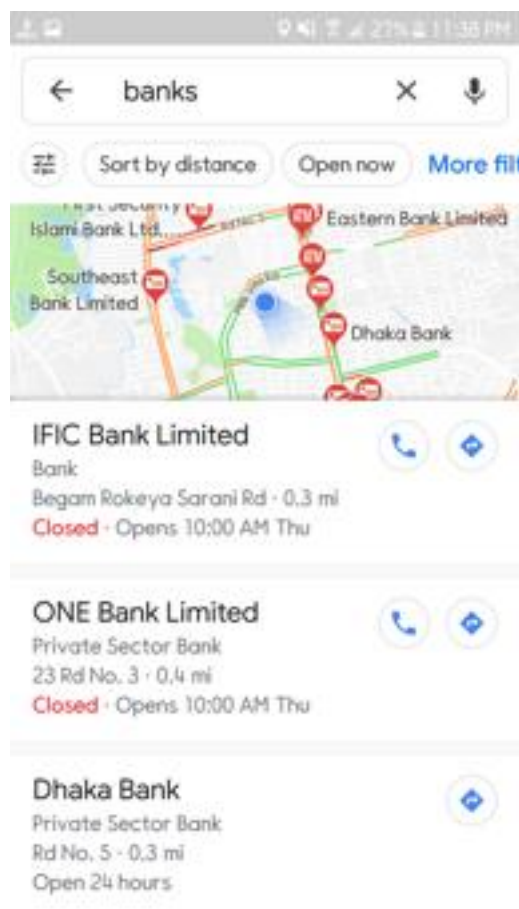


Figure 5.14: Nearby Banks

5.4.8.2. Nearby Hospitals

Navigation map view for nearby hospitals.

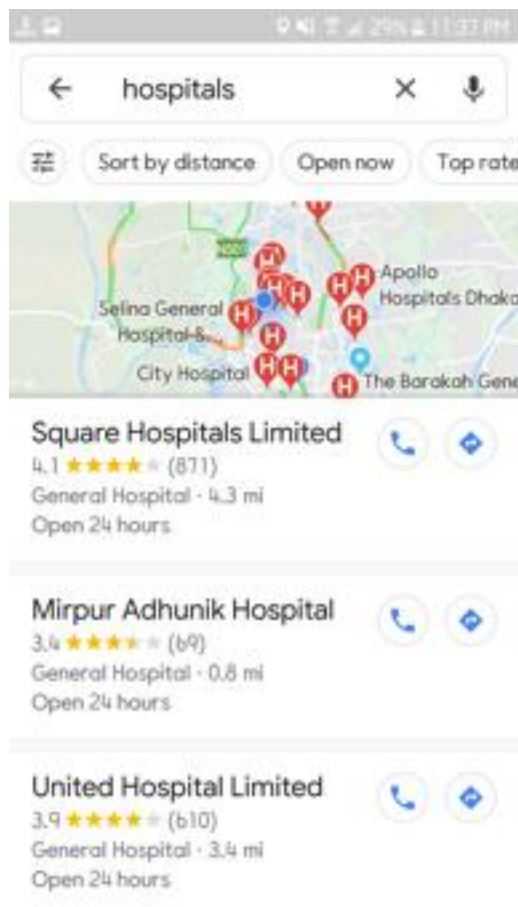


Figure 5.15: Nearby Hospitals

5.4.8.3 Nearby Police station

Navigation map view for nearby police station.

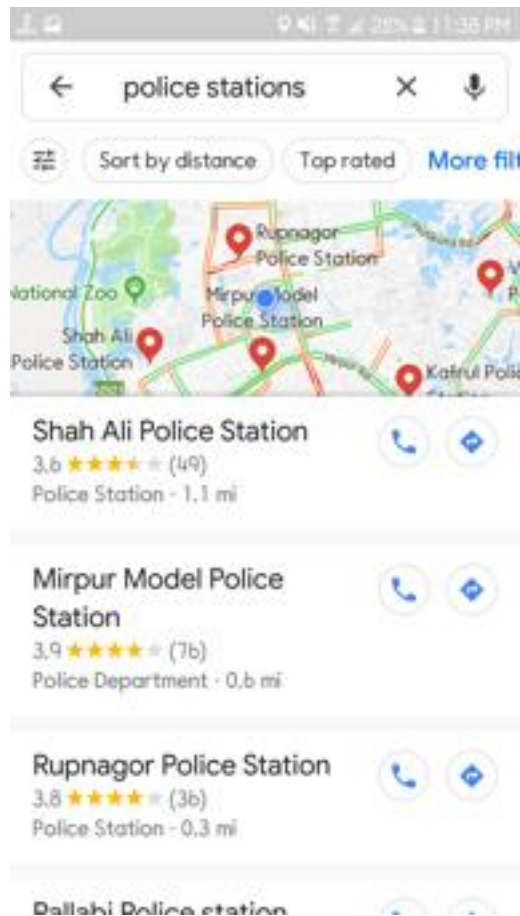


Figure 5.16: Nearby Police station

5.4.8.4. Nearby Restaurant

Navigation map view for nearby restaurants.



Figure 5.17: Nearby Restaurant

5.4.8 Weather Forecast

This feature contains search city wise weather forecast for five days. And each day contains eight data. Google does not provided all place weather forecast of Bangladesh just for major cities are provided. By selecting each category users get five days weather forecast like below figure.



Figure 5.18: Weather Forecast

CHAPTER 6

Testing

6.1 Testing Implementation

Let us first understand what implementation means. Implementation is the process of putting an action for the formulated plan. Before we implement, the plan should have been completed and our objectives should be clear.

Testing each one of those actions formulated in the plan is said to be implementation testing.

After unit testing is completed, developer performs integration testing. It is the process of verifying the interfaces and interaction between modules. While integrating, there are lots of techniques used by developers and one of them is the incremental approach.

In Incremental integration testing, the developers integrate the modules one by one using stubs or drivers to uncover the defects. This approach is known as incremental integration testing. To the contrary, big bang is one other integration testing technique, where all the modules are integrated in one shot.

Incremental Testing Methodologies

- Top down Integration - This type of integration testing takes place from top to bottom. Unavailable Components or systems are substituted by stubs
- Bottom Up Integration - This type of integration testing takes place from bottom to top. Unavailable Components or systems are substituted by Drivers
- Functional incremental - The Integration and testing takes place on the basis of the functions or functionalities as per the functional specification document.

- Incremental Testing - Features
- Each Module provides a definitive role to play in the project/product structure
- Each Module has clearly defined dependencies some of which can be known only at the runtime.
- The incremental integration testing's greater advantage is that the defects are found early in a smaller assembly when it is relatively easy to detect the root cause of the same.
- A disadvantage is that it can be time-consuming since stubs and drivers have to be developed for performing these tests. Another disadvantage is user needs data connection and location on to their devices.

The configuration of the tester machines were as follows:

Developers PC:

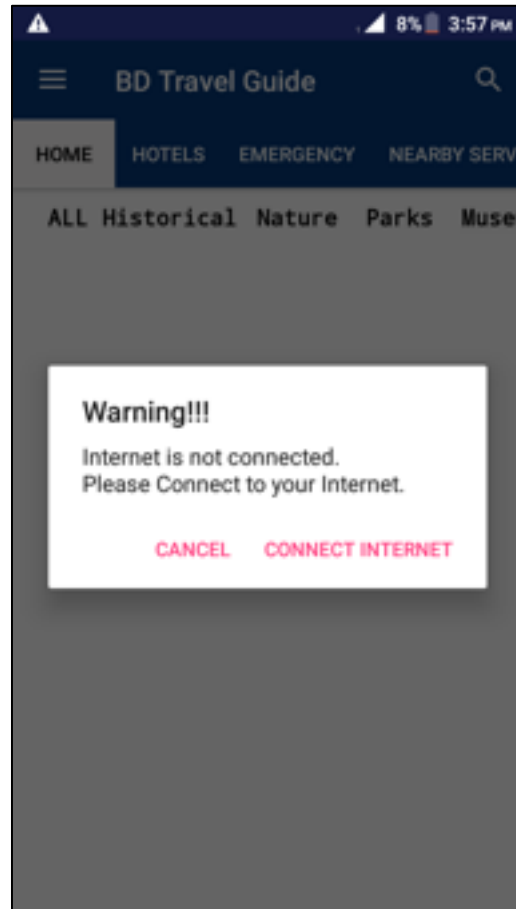
Windows Edition	Windows 10 Home
Installed Memory(RAM)	8.00GB DDR3 RAM
Processor	Intel Core i7 Processor (6th Gen)
System Type	64 bit Windows 10 Operating System

Mobile Device 1:

Model	Samsung Galaxy S6
Processor	1.5GHz octa-core
Operating System	Android 5.0
Memory	RAM 3GB, Internal storage 32GB
DISPLAY	Capacitive Touchscreen

Internet testing:

When user uses application offline this alert message will show. So user need to connect their internet.



Location testing:

When user wants to use nearby services they need to provide their device location either this alert message will show to enable location.



6.2 Test Results and Reports

Travel list, hotels, nearby services and weather forecast tables are used to obtain details about how the tests are organized in the system. All the testing result and report, already gathered to know the application advance work which will be developed in future.

SL.NO	Objects	Results
1	To check if the internet connection is on or not. Also make sure location is on for utilize nearby services. By this loaded all the server data.	Successful
2	To check if the internet connection is on or not. Also make sure location is on for utilize nearby services. By this loaded all the server data.	Successful
3	To check all features are loaded or not in the Home page of main view.	Successful
4	To check if map feature like nearby hospitals, police stations, bank shows the information or not.	Successful
5	To check “weather forecast” option work or not Successful	Successful

Table 5.15: Test Results and Reports

6.3 Important code snippet of the system

ApiInterface.java to parse data from live server

```
import com.jesminnipu.touristguide.model.Response;

import java.util.ArrayList;
import java.util.List;

import retrofit2.Call;
import retrofit2.http.GET;
import retrofit2.http.Query;

public interface ApiInterface {

    @GET("/divisions.php")
    Call<ArrayList<Response>> getDivname();

    @GET("/places.php")
    Call<ArrayList<DivInformation>> getdivInfo(@Query("type") String type, @Query("division")

    @GET("/place.php")
    Call<List<PlaceInformation>>getPlace(@Query("id") int id);

    @GET("/hotels.php")
    Call<ArrayList<Hotel>> getHotels(@Query("division") String division);

    @GET("/hotel.php")
    Call<List<Hotel>> getHotel(@Query("id") int id);

}
```

Build.gradle for REST API uses for map and weather forecast

```
buildTypes {
    release {
        minifyEnabled false
        proguardFiles getDefaultProguardFile('proguard-android.txt'), 'proguard-rules.pro'
    }
}
productFlavors {
}

dependencies {
    implementation fileTree(include: ['*.jar'], dir: 'libs')
    androidTestImplementation('com.android.support.test.espresso:espresso-core:2.2.2', {
        exclude group: 'com.android.support', module: 'support-annotations'
    })
    implementation 'com.google.code.gson:gson:2.3.1'
    implementation 'com.android.support:appcompat-v7:27.1.1'
    implementation 'com.android.support:design:27.+'
    implementation 'com.squareup.retrofit2:retrofit:2.3.0'
    implementation 'com.squareup.retrofit2:converter-gson:2.3.0'
    implementation 'com.android.support.constraint:constraint-layout:1.0.2'
    implementation 'com.android.support:recyclerview-v7:27.+'
    implementation 'com.android.support:cardview-v7:27.+'
    implementation 'com.github.bumptech.glide:glide:3.7.0'
    implementation 'com.android.support:support-v4:27.+'
    testImplementation 'junit:junit:4.12'
    implementation 'com.squareup.okhttp3:logging-interceptor:3.9.0'
    // implementation 'io.nlopez.smartlocation:library:3.3.3'
    implementation 'com.google.android.gms:play-services-location:15.0.1'
    implementation 'com.squareup.picasso:picasso:2.5.0'
    implementation files('libs/vip.parvez.unixtogmt.jar')
    implementation 'com.github.pwittchen:reactivenetwork-rx2:1.0.0'
}
```

AndroidManifest.xml For network use permissions

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.jesminnipu.touristguide">

    <uses-permission android:name="android.permission.INTERNET" />
    <uses-permission android:name="android.permission.ACCESS_NETWORK_STATE" />
    <uses-permission android:name="android.permission.CALL_PHONE" />
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION" />
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/AppTheme">
        <activity
            android:name=".activity.MainActivity"
            android:label="@string/app_name"
            android:theme="@style/AppTheme.NoActionBar">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        <activity android:name=".activity.PlaceDetailsActivity" />
        <activity android:name=".activity.HotelDetailsActivity"></activity>
    </application>
```

MyService.java for showing Alert dialog Message

```
import android.content.Context;
import android.content.DialogInterface;
import android.support.v7.app.AlertDialog;
import android.util.Log;

/**
 * Created by NIPU on 1/14/2018.
 */

public class MyService {

    /**
     * @param msg
     */
    public static void L(String msg) { Log.d( tag: "JESMIN::", msg); }
    public static ProgressDialog progressDialog(Context context, String msg) {
        ProgressDialog progressDialog = new ProgressDialog(context);
        progressDialog.setMessage(msg);
        progressDialog.setCancelable(false);
        return progressDialog;
    }

    public static boolean alertDialog(Context context, String msg) {
        AlertDialog alertDialog = new AlertDialog.Builder(context).create();
        alertDialog.setTitle("Warning!!");
        alertDialog.setMessage(msg);

        alertDialog.setButton(AlertDialog.BUTTON_NEUTRAL, text: "OK",
            (OnClickListener) (dialog, which) → { dialog.dismiss(); });
        alertDialog.show();
        return false;
    }
}
```

CHAPTER 7

Conclusion and future scope

7.1 Discussion & Conclusion

We know Bangladesh is an underdeveloped country but most people of our country use internet Now-a-days. There was a time when people wanted to go for vacation in some tourist places, they did not know how to go these places. For that reason they had to suffer many problems about visiting that places. They had also problems for choosing the best tourist places or hotels.

We are living in the era of modern technology. Nowadays everything is based on Computer science and information technology. Today the world is smaller than yesterday. IT sector is one of the most critical but with great potential revenue source. Using Internet is becoming a habit. According to Internet in 2015 23.8% people use smart phone in Bangladesh. Surely in coming years this number will get bigger. This android application designed so that people don't have to invest huge time for getting information about a tour plan or an exhibition. Users can enjoy these features by using their phone. Initiated this project as an android application, with that hope people will get desired services.

7.2 Scope for Future Career

In this android app people can use it for touring places information, hotels information, nearby humanities such as hospitals, restaurants, police stations and banks, emergency calling system and weather forecast system for five days system. There are huge amount of traveling places in our country. In future will add more travel places information day by day when get available, also want to link up other features of Google map so users can easily find places destination, path duration etc. For real time system, this app used open weather API and google elevation map API, as free googles API has usage limitations, these API does not can make API call frequently. But in future, will provided rest of information in the server and buy API as user requirements when this app have great amount and user response.

Appendices

Appendix B: Related Diagrams

In this section given most the php code snippet.

B.1 DATABASE CONNECTION

```
<?php

$servername = "localhost";
$username = "root";
$password = "";
$database_name = "tourim";

// Create connection
$conn = new mysqli($servername, $username, $password, $database_name);
// Check connection
if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
} else {
}

$conn->set_charset("utf8");
?>
```

Figure B1: Database Connection

B.2 DATABASE HELPER CLASS

```
<?php

include 'database.php';

class DatabaseHelper {

    public function getPlaces( $type, $division ) {
        $sql = "";
        if ( isset( $type ) && isset( $division ) ) {
            $sql = "SELECT * FROM places WHERE division_name = '$division' AND type = '$type'";
        } elseif ( isset( $type ) ) {
            $sql = "SELECT * FROM places WHERE type = '$type'";
        } elseif ( isset( $division ) ) {
            $sql = "SELECT * FROM places WHERE division_name = '$division'";
        } else {
            $sql = "SELECT * FROM places";
        }

        $result = $GLOBALS['connection']->query( $sql );
        if ( $result->num_rows > 0 ) {
            return $result;
        } else {
            return false;
        }
    }
}
```

Figure B.2: Helper Class

B.3 Tour Places

```
<?php
require 'helper/database_helper.php';
header('Content-type: application/json');

$resultArray = array();
$dbHelper = new DatabaseHelper();

$http = isset($_SERVER['HTTPS']) && $_SERVER['HTTPS'] == 'on' ? "https://" : "http://";
$url = $http . $_SERVER["SERVER_NAME"] . "/tourim/images/" . $_SERVER['REQUEST_URI'];

if (isset($_GET['type']) && isset($_GET['division'])) {
    $result = $dbHelper->getPlaces($_GET['type'], $_GET['division']);
} elseif (isset($_GET['type'])) {
    $result = $dbHelper->getPlaces($_GET['type'], null);
} elseif (isset($_GET['division'])) {
    $result = $dbHelper->getPlaces(null, $_GET['division']);
} else {
    $result = $dbHelper->getPlaces(null, null);
}
```

Figure B3: Tour Place.

B4. Hotels

```
<?php
require 'helper/database_helper.php';
header('Content-type: application/json');

$resultArray = array();
$dbHelper = new DatabaseHelper();

$http = isset($_SERVER['HTTPS']) && $_SERVER['HTTPS'] == 'on' ? "https://" : "http://";
$url = $http . $_SERVER["SERVER_NAME"] . "/tourim/images/" . $_SERVER['REQUEST_URI'];

if (isset($_GET['id'])) {
    $result = $dbHelper->getHotel($_GET['id']);

    if ($result) {
        while ($row = $result->fetch_assoc()) {
            $array = array();
            $array['id'] = $row['id'];
            $array['title'] = $row['title'];
            $array['address'] = $row['address'];
            $array['contact'] = $row['contact'];
            $array['description'] = $row['description'];
            $array['image'] = $row['image'];
            $array['division_name'] = $row['division_name'];
            $array['price_type'] = $row['price_type'];
            array_push($resultArray, $array);
        }
    }
}
```

Figure B4: Hotels

References

- [1] Travel Bangladesh available at
<https://play.google.com/store/apps/details?id=app.marszone.travelguide>
- [2] Weather for Bangladesh available at
<https://play.google.com/store/apps/details?id=com.ssaurel.bangladeshweather>.
- [3] Travel Bangladesh available at
https://play.google.com/store/apps/details?id=com.j.travel_bd.
- [4] Hotels of Bangladesh available at
<https://play.google.com/store/apps/details?id=com.shamimnasir.bdhotels>
- [5] Place list <https://www.touristplaces.com.bd/>
- [6] Place details information <http://visitbangladesh.gov.bd/>
- [7] For place information <https://www.tripadvisor.com/>
- [8] Hotels information available at <https://www.tripadvisor.com/>
- [9] XML available at <https://www.w3schools.com/xml/default.asp>.
- [10] JAVA available at https://www.java.com/en/download/whatis_java.jsp.
- [11] PHP available at www.w3schools.com/php/default.asp.
- [12] MySQL available at https://www.w3schools.com/sql/sql_intro.asp.
- [13] Weather Forecast API available at <https://openweathermap.org/forecast16>